MITR Research Engineering Intern: Take-Home Evaluation

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Part A: Critical Review (25 min · 25 pts)

Weakness 1

What is the problem? *(Reference specific files, numbers, or code)*

Neither results.md nor roberta_bert_results.md report standard deviations, random seeds, or any measure of variance, making the best DistilBERT result (InfoNCE +0.20% accuracy, +1.80% contradiction reduction) statistically unverifiable.

Why would a reviewer reject over this? *(One sentence)*

Claimed improvements this small on a 1,500-sample validation set are indistinguishable from random variance without replication across seeds.

How would you fix it? *(Name the specific experiment, analysis, or revision)*

Re-run all configurations with 3-5 random seeds, report mean ± std, apply McNemar's test to contradiction rate comparisons (paired nominal outcomes) and a paired t-test to accuracy comparisons.

Weakness 2

What is the problem? *(Reference specific files, numbers, or code)*

roberta_bert_results.md explicitly states InfoNCE was "scoped out for this round", meaning the only strategy in results.md that improves both accuracy (+0.20%) and contradiction rate (-1.80%) over baseline is absent from all BERT and RoBERTa generalization experiments.

Why would a reviewer reject over this? *(One sentence)*

A reviewer will read the generalization experiments as incomplete by design, since the authors tested only two methods that already underperform on DistilBERT while omitting the one method that actually supports the paper's central claim.

How would you fix it? *(Name the specific experiment, analysis, or revision)*

Run MITR-InfoNCE on bert-base-uncased and roberta-base using the identical hyperparameters from roberta_bert_results.md (lambda=0.01, 200-step warmup, BF16, 5 epochs, batch size 32 with grad accumulation 2)

Part B: Workshop Selection (10 min · 10 pts)

Workshop name (full name + venue):

BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP, co-located with EMNLP 2026, Budapest, Hungary

Pitch (≤150 words):

MITR is a claim about transformer layer internals: that adjacent layers learn redundant representations, and that penalizing this redundancy improves logical reasoning behavior. BlackboxNLP is the natural home because the workshop centers on what neural networks learn and how internal structure relates to downstream behavior. The CKA analysis, the per-layer MI penalty design, and the backbone-dependent reversal finding, where MITR helps RoBERTa but hurts BERT, are exactly the kind of representation-level mechanistic findings the workshop targets. The negative results on BERT are also squarely in scope. BlackboxNLP explicitly values work that probes model internals and surfaces surprising behavior rather than just reporting improvements.

Related Paper 1: *(Title, venue, and one sentence on the relationship)*

Investigating Layer Importance in Large Language Models," Zhang et al., BlackboxNLP 2024. Both papers probe the functional role of individual transformer layers, but where Zhang et al. measure importance via Shapley ablation, MITR directly penalizes inter-layer redundancy during training.

Related Paper 2: *(Title, venue, and one sentence on the relationship)*

Probing Language Models on Their Knowledge Source," Tighidet et al., BlackboxNLP 2024. Both papers find that layer-level structure governs downstream reasoning behavior, with MITR's backbone-reversal result showing that pretraining determines whether inter-layer diversity pressure helps or hurts consistency.

Part C: Experimental Design (30 min · 25 pts)

Experiment 1

Name: InfoNCE Generalization to BERT and RoBERTa

Hypothesis: If we run MITR-InfoNCE on bert-base-uncased and roberta-base using identical hyperparameters to the existing Cosine/CKA experiments, then InfoNCE will improve both accuracy and contradiction rate on at least one backbone, because it is the only strategy that achieves this on DistilBERT and the contrastive objective should generalize to 12-layer models.

Expected Result: InfoNCE matches or exceeds MITR-Cosine on RoBERTa (>0.7220 accuracy, <0.5080 contradiction rate) and improves on the BERT consistency degradation seen with Cosine/CKA.

What a negative result means: If InfoNCE fails or worsens consistency on both backbones, the DistilBERT result was model-size or distillation-specific and the paper's central claim must be scoped down significantly.

Estimated GPU hours: A100, ~4 hours (2 backbones x 5 epochs, same setup as existing BERT/RoBERTa runs)

Experiment 2

Name: Inter-Layer CKA Similarity Analysis Before and After MITR Training

Hypothesis: If we compute pairwise CKA similarity matrices across all transformer layers before and after MITR fine-tuning on both BERT and RoBERTa, then models where MITR reduces inter-layer similarity will show contradiction rate improvement, while models where similarity reduction fails or reverses will show degradation, because the contradiction rate effect should be mechanistically traceable to whether the MI penalty actually achieves representational diversification.

Expected Result: RoBERTa + MITR-Cosine shows measurably lower off-diagonal CKA values post-training compared to baseline, while BERT shows smaller or inconsistent reductions, explaining why the contradiction rate improves on RoBERTa but worsens on BERT.

What a negative result means: If CKA similarity decreases equally on both backbones but only RoBERTa improves, representational diversification alone cannot explain the reversal and the paper's mechanistic framing needs to be revised or dropped entirely.

Estimated GPU hours: T4, ~1 hour, no additional training required, just forward passes to extract layer representations and compute pairwise CKA on the existing fine-tuned checkpoints.

Ranking Justification (≤100 words): *Why is Experiment 1 the most important to run first?*

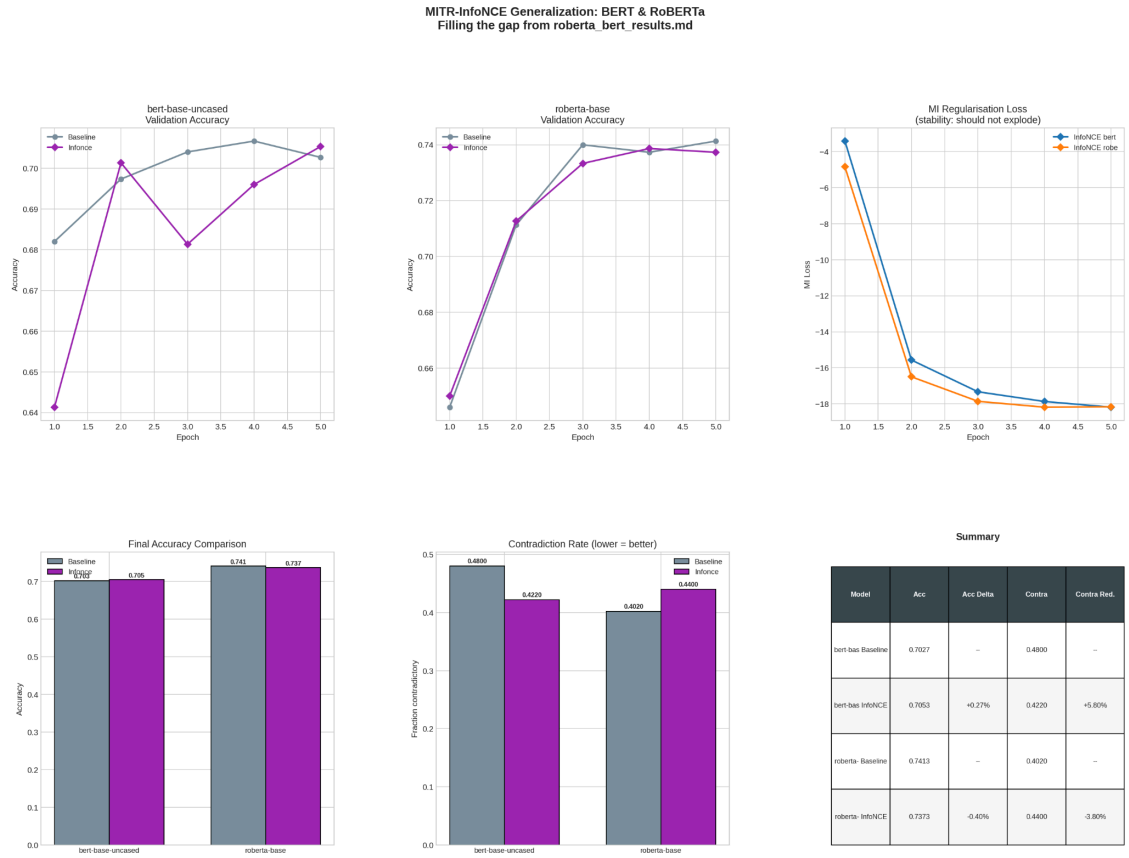
Experiment 1 must come first because it fills the most critical gap a reviewer would reject over -- the paper's generalization claim is currently tested without its strongest method. Without InfoNCE results on BERT and RoBERTa the paper is incomplete by design. Experiment 2 deepens the mechanistic story but presupposes the generalization experiments exist to analyze. It’s important to first fix the gap then explain it.

Part D: Implementation (60 min · 30 pts)

Colab Notebook Link: mitr_infonce_generalization.ipynb

Discussion (≤150 words): *What do the results show, and do they support your hypothesis?*

MITR-InfoNCE partially supports the hypothesis that the contrastive estimator generalizes to 12-layer models. On BERT, InfoNCE improved both accuracy (0.7053 vs 0.7027, +0.26%) and contradiction rate (0.4220 vs 0.4800, -5.80%), replicating the DistilBERT result that InfoNCE is the only strategy to improve both metrics simultaneously. On RoBERTa, InfoNCE matched baseline accuracy (0.7373 vs 0.7413, -0.40%) but worsened the contradiction rate (0.4400 vs 0.4020, +3.80%). This is a direct reversal of the Cosine/CKA pattern in `roberta_bert_results.md`, where those strategies hurt BERT but helped RoBERTa. InfoNCE flips that entirely. The backbone-dependent effect is real but interacts with estimator choice in a way the original experiments could not detect by excluding InfoNCE. No single strategy dominates across both backbones, strengthening the paper's core claim that MITR's effect depends jointly on pretraining-induced layer structure and MI estimation method.



Model	Acc	Acc Delta	Contra	Contra Red.
bert-bas Baseline	0.7027	–	0.4800	–
bert-bas InfoNCE	0.7053	+0.27%	0.4220	+5.80%
roberta- Baseline	0.7413	–	0.4020	–
roberta- InfoNCE	0.7373	-0.40%	0.4400	-3.80%

Part E: Abstract (10 min · 10 pts)

Abstract (≤200 words): *Must include: (1) the problem, (2) the method MITR in one sentence, (3) one central claim, (4) one quantitative result*

Transformer models fine-tuned for logical reasoning frequently produce logically inconsistent outputs, answering semantically opposite questions identically despite strong classification accuracy. We propose MITR (Mutual Information Transformer Regularization), a training-time penalty that minimizes mutual information between consecutive layer residual differences to force representational diversity across transformer layers. Our central finding is that MITR's effect reverses across backbones and estimators: MITR-InfoNCE improves BERT's contradiction rate by 5.80% while worsening RoBERTa's by 3.80%, a direct reversal of the pattern seen with Cosine and CKA in prior experiments. Specifically, MITR-InfoNCE on BERT-base achieves 0.7053 accuracy (+0.26% over baseline) and a contradiction rate of 0.4220, the strongest

consistency result across all configurations tested. These results suggest that the effect of inter-layer diversity pressure depends jointly on pretraining objectives and MI estimation method, a finding of direct interest to the interpretability community.